

Healthcare Management Portal

Product Requirements Document (PRD)

Overview

HealthConnect is a web-based healthcare platform enabling patients, doctors, hospital staff, laboratories, pharmacists, and administrators to manage appointments, medical records, prescriptions, billing, and communication securely.

Business Goals

Improve patient engagement, digitize records, streamline appointments, support telemedicine, improve operational efficiency, and ensure HIPAA/GDPR-aligned security.

User Roles

Patient, Doctor, Receptionist, Laboratory Staff, Pharmacist, Hospital Admin, Super Admin.

Core Modules

Registration, Login/MFA, Patient Dashboard, Doctor Dashboard, Appointment Management, Medical Records, Prescription Management, Laboratory, Billing, Notifications, Telemedicine, Pharmacy, Reports & Analytics.

Functional Highlights

Unique email/phone, OTP verification, appointment booking/rescheduling, electronic prescriptions, lab workflow, online payments, notifications via Email/SMS/Push.

Non-Functional Requirements

Response time <2s, 99.9% uptime, HTTPS, RBAC, MFA, encryption, audit logs, scalability for multi-hospital deployments, WCAG accessibility.

Integrations

Payment Gateway, SMS, Email, Video Consultation API, Insurance API, Laboratory API, Pharmacy API.

Database Entities

Patient, Doctor, Appointment, Prescription, Medical Record, Invoice, Payment, Notification, Laboratory Test, Laboratory Report.

KPIs

Appointment Success >98%, Registration >99%, Payment >99%, Response Time <2 sec, Availability 99.9%, Patient Satisfaction >4.5/5.

Future Enhancements

AI symptom checker, AI appointment assistant, wearable integrations, AI prescription validation, AI report summarization, predictive analytics.